

**ALIGN-VQA: ALIGNMENT-GUIDED RESIDUAL MODELING
FOR LONGITUDINAL MEDICAL VISUAL QUESTION
ANSWERING**

by

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AUTHOR'S DECLARATION

I, Kaung Sithu, declare that the research work carried out for this thesis was in accordance with the regulations of the Asian Institute of Technology. The work presented in it are my own and has been generated by me as the result of my own original research, and if external sources were used, such sources have been cited. It is original and has not been submitted to any other institution to obtain another degree or qualification. This is a true copy of the thesis including final revisions.

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ABSTRACT

Longitudinal image comparison is fundamental to radiological decision-making, requiring clinicians to assess the temporal evolution of findings across visits. Medical Difference Visual Question Answering (Diff-VQA) formalizes this process, however, the current methods for Diff-VQA often rely on direct feature subtraction or simple fusion. These strategies assume inherent spatial correspondence, a premise frequently violated by stochastic acquisition variability and non-rigid anatomical deformations in clinical practice. The author proposes ALIGN-VQA, a novel alignment-guided residual modeling architecture for robust temporal reasoning. This approach introduces three key innovations: (i) Cross-Image Difference Attention (CIDA): Aligns prior-visit features to the current scan via semantic reconstruction, ensuring misalignment-tolerant comparison, (ii) Learnable Residual Gating: Captures precise deviations between aligned historical and current representations to extract evolution-aware features, (iii) Question-Guided Difference Tokenizer (QDT): Selectively pools question-relevant temporal changes for targeted reasoning. Evaluation on the Medical-Diff-VQA benchmark, ALIGN-VQA shows comparable or better performance compared to state-of-the-art methods, with notable improvements in complex reasoning categories. Further, the proposed model ALIGN-VQA is a lightweight architecture making it amenable for edge deployments.

CONTENTS

	Page
TITLE PAGE	i
AUTHOR’S DECLARATION	iii
ACKNOWLEDGEMENTS	iv
ABSTRACT	v
LIST OF TABLES	viii
LIST OF FIGURES	ix
CHAPTER 1 INTRODUCTION	1
1.1 Background of the Study	1
1.2 Statement of the Problem	1
1.3 Research Questions	2
1.4 Hypotheses	2
1.5 Objectives of the Study	2
1.6 Methodology	3
1.7 Scope	3
1.8 Expected Key Findings	4
1.9 Contribution	4
CHAPTER 2 RELATED WORK	5
2.1 Difference Modeling and the Alignment Problem in Longitudinal VQA	5
2.2 Feature Selection and Computational Efficiency	6
2.3 Mitigating Hallucination and Dataset Biases through Counterfactual Training	6
2.4 Hypothesis and Positioning	7
2.5 Comparative Analysis of Methodologies	7
CHAPTER 3 METHODOLOGY	9
3.1 Architectural Overview	9
3.2 Problem Formulation	10
3.3 Architectural Components	10
3.3.1 Visual Feature Extraction	10
3.3.2 Cross Image Difference Attention	10

3.3.3	Learnable Gated Residual Modeling	11
3.3.4	Domain-Aware Text Encoding	11
3.3.5	Question-Guided Difference Tokenization	11
3.3.6	Conditional Answer Generation	12
3.3.7	Counterfactual Regularizer	12
3.3.8	Masked Residual Modeling	12
3.4	Multi-Stage Training Strategy	12
3.5	Implementation Details	14
CHAPTER 4	EXPERIMENTAL RESULTS	15
4.1	Rationale	15
4.2	Dataset	15
4.2.1	Pathological Distribution	16
4.2.2	Question Type Distribution	17
4.2.3	Linguistic Distribution	17
4.3	Experiment Setting	18
4.3.1	Preprocessing	18
4.3.2	Models and Techniques	18
4.3.3	Training Procedure	19
4.3.4	Evaluation Metrics	20
4.4	Quantitative Results on the MIMIC-Diff-VQA Benchmark	20
4.5	Ablation Studies and Component Analysis	21
4.5.1	Impact of Architectural Modules	21
4.5.2	Impact of Token Sparsity	21
REFERENCES		23
APPENDICES		26
VITA		27

LIST OF TABLES

Tables	Page
Table 2.1 Comparative Analysis of Limitations in State-of-the-Art DiffVQA Models.	8
Table 2.2 Mapping of Proposed Components to Research Gaps	8
Table 4.1 Quantitative performance comparison of the proposed model against state-of-the-art methods on the MIMIC-Diff-VQA dataset.	22
Table 4.2 Ablation studies isolating the impact of the Top- k selection threshold, Cross-Image Difference Attention (CIDA), and Counterfactual (CF) regularizer.	22

LIST OF FIGURES

Figures	Page
Figure 3.1 High Level Architecture Overview	9
Figure 3.2 Masked Residual Modeling	13
Figure 4.1 Prevalance of Medical Findings	16
Figure 4.2 Distribution of Question Types	17
Figure 4.3 Distribution of Question Lengths	18

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The automated interpretation of longitudinal radiological images, a cornerstone of clinical practice, has recently been formalized under the task of Difference Visual Question Answering (DiffVQA). The release of Hu et al. (2025) was a pivotal moment, enabling the development of models to answer nuanced questions about temporal changes in patient scans. The subsequent evolution of state-of-the-art models has revealed a consistent, yet resource-intensive, paradigm.

Initial architectural innovations, such as the graph-based representations in Hu et al. (2023) and the dedicated residual encoder in Lu et al. (2024), established methods for processing paired image features. However, these models operate on dense visual feature maps, a computationally expensive approach. Recognizing the need for stronger representations, Cho, Kim, Shin, Cho, and Shin (2024) demonstrated that extensive pre-training on large-scale natural and medical corpora significantly boosts performance, but at the cost of being highly data-hungry and dependent on external reports. More recently, retrieval-augmented models like Yung et al. (2024) and powerful backbones like the Swin Transformer in the Marhuenda, Obrador-Reina, Aas-Alas, Albiol, and Paredes (2025) have pushed performance further. While these models are highly effective, they represent a "brute-force" methodology: they rely on massive model scale, computationally expensive database searches, or the processing of thousands of visual tokens from the entire image to implicitly learn the difference signal.

1.2 Statement of the Problem

The dominant paradigm in Difference VQA is fundamentally inefficient and lacks explicit clinical reasoning. Current models are computationally expensive, requiring the processing of dense visual information from two full images, even when the question pertains to a small, localized change. This brute-force approach is not only wasteful but also fails to explicitly model the directionality of clinical findings, making it difficult to distinguish between concepts like "worsened" versus "improved." Furthermore, this reliance on large, complex models without strong logical constraints makes them prone to hallucination—confidently reporting changes that do not exist, a critical failure for any

clinical support tool. The field lacks a model that is computationally efficient, explicitly models the direction of change, and is robust against hallucination.

1.3 Research Questions

Following the problem, There are three research questions:

1. Can longitudinal, difference-focused questions be accurately answered using only a handful of question-guided difference tokens, instead of processing all image patches?
2. Does the cross image difference attention help the model distinguish the direction of change such as missing or new abnormalities?
3. Can a counterfactual training signal, such as swapping images or negating terms in the question, reduce the model’s tendency to hallucinate differences?

1.4 Hypotheses

There are three hypotheses to answer the research question:

1. A small set of sparse, question-guided difference tokens is sufficient to achieve state-of-the-art performance, leading to a dramatic increase in computational efficiency.
2. Aligning reference image to current image via CIDA yields more accurate representation of disease evolution than direct substraction.
3. Training with a counterfactual regularizer will result in a lower false positive rate on unchanged image pairs and improved logical consistency compared to models trained without it.

1.5 Objectives of the Study

The main objective is to design and develop lightweight, difference aware medical VQA system that can provide faithful visual evidence to support its answer while being compute efficient and not requiring retrieval databases, radiology reports or expert annotations. The sub-objectives are:

1. To validate the effectiveness of a question-guided tokenizer (QDT) at distilling visual information into a sparse set of tokens.
2. To evaluate the impact of an integrated cross image difference attention (CIDA) on the model’s ability to reason about the direction of clinical change.
3. To measure the reduction in model hallucination achieved by introducing a counterfactual (CF) regularizer.

1.6 Methodology

This work introduces ALIGN-VQA, a novel, lightweight architecture that shifts from a brute-force comparison to an alignment-guided residual modeling paradigm. The methodology is centered on three key innovations designed to handle the stochastic acquisition variability and non-rigid anatomical deformations inherent in longitudinal imaging. First, to overcome the limitations of spatial misalignment between scans, cross image difference attention (CIDA) module semantically aligns prior-visit features to the current scan via cross-attention before computing differences. This aligned representation is then passed through a Learnable Residual Gating mechanism, which adaptively suppresses acquisition noise to extract precise, evolution-aware residual features. Second, a Question-Guided Difference Tokenizer (QDT) utilizes domain-aware text embeddings from ClinicalBERT to evaluate the relevance of these visual deviations by applying cross-attention to selectively pool only the Top-K most clinically salient temporal changes. The vision components learn contextual coherence via a self-supervised Masked Residual Modeling (MRM) task, while an auxiliary multi-label classification head explicitly ground the visual features in 14 thoracic pathologies.

1.7 Scope

In scope:

- The thesis will exclusively use the public Medical-Diff-VQA dataset, focusing on all question categories for all experiments.
- The core of the work is the design, implementation, and evaluation of our novel, lightweight VQA architecture and its specific modules (CIDA, QDT, CFA-Reg, MRM).
- Evaluation will consist of targeted ablation studies and a direct benchmark against a comprehensive suite of SOTA models (EKAID, ReAI, PLURAL, RegioMix, VED).

Out of scope:

- This work will not involve the collection of new clinical data or the execution of clinical trials with human radiologists.
- This thesis will not develop a new, general-purpose medical VQA system.

1.8 Expected Key Findings

- The proposed, ALIGN-VQA, model is expected to achieve performance comparable or superior to current state-of-the-art models with a fraction of the computational cost, demonstrating the viability of an efficiency-focused paradigm.
- The integration of Cross-Image Difference Attention (CIDA) and Learnable Residual Gating will be shown to provide a significant accuracy boost by explicitly handling spatial misalignments and suppressing stochastic acquisition noise, particularly excelling in complex comparative reasoning categories.
- The Counterfactual Regularizer will be proven to significantly disentangle true pathological evolution from dataset biases, demonstrating a measurable improvement in model robustness and a reduction in spurious correlations.

1.9 Contribution

- This thesis introduces ALIGN-VQA, a novel, lightweight paradigm for DiffVQA centered on efficiency, explicit reasoning, and robustness, shifting the field away from the dominant brute-force, resource-intensive methodologies.
- Three key architectural innovations are developed: Cross-Image Difference Attention (CIDA) for misalignment-tolerant semantic reconstruction, Learnable Residual Gating for isolating true evolution-aware features, and a Question-Guided Difference Tokenizer (QDT) that selectively pools clinically salient tokens to prevent signal dilution and maximize computational efficiency.
- A comprehensive multi-stage training strategy that incorporates self-supervised Masked Residual Modeling (MRM), auxiliary disease classification, and counterfactual learning—is introduced. This yields a state-of-the-art framework that is robust, computationally accessible, and self-contained, removing the reliance on massive external radiology report corpora for pre-training.

CHAPTER 2

RELATED WORK

This chapter critically examines the existing body of literature to contextualize ALIGN-VQA architecture and the research questions driving this thesis. Longitudinal medical imaging is a cornerstone of radiological diagnosis, requiring clinicians to meticulously track disease progression or stability over time. In response, Difference-aware Medical VQA (Diff-VQA) has emerged as a specialized domain designed to analyze paired longitudinal scans and answer complex clinical queries regarding their differences. However, despite rapid advancements in generative vision-language models, current Diff-VQA architectures remain constrained by several critical methodological bottlenecks.

2.1 Difference Modeling and the Alignment Problem in Longitudinal VQA

Early approaches to this problem, such as Hu et al. (2023), attempt to structure the comparison by representing anatomical regions as nodes in a multi-relationship graph. By extracting features from identical anatomical bounding boxes across two temporal scans, it aims to evaluate disease progression within specific lung regions. While this explicitly incorporates spatial and semantic medical priors, its reliance on a pre-trained Ren, He, Girshick, and Sun (2015) to detect bounding boxes leaves the model vulnerable; inaccurate feature extraction directly results in incorrect comparative predictions. Furthermore, this strict graph structure struggles to adapt to non-rigid anatomical deformations or novel structural changes not predefined in the knowledge graph.

More recent models, such as Lu et al. (2024), compute temporal differences by directly subtracting the reference image from the main image to produce a residual input. This residual is processed through an independent encoder, guided by a consistency loss to enforce focus on disparate regions. However, direct subtraction strategies implicitly assume perfect spatial correspondence between the two scans. In clinical practice, this assumption is frequently violated due to stochastic acquisition variability, patient movement, and postural differences between visits. When raw image subtraction is applied to misaligned scans, it generates overwhelming artifact noise rather than isolating true pathological evolution.

2.2 Feature Selection and Computational Efficiency

Cho et al. (2024) created addresses longitudinal VQA by concatenating the flattened feature vectors of both the past and current images into a single, massive sequence. Utilizing He, Zhang, Ren, and Sun (2016), it generates 576 tokens per image, resulting in a dense, highly redundant input matrix that is processed by a 6-layer Transformer encoder-decoder. While PLURAL benefits from large-scale pretraining on external datasets like Johnson, Pollard, Mark, Berkowitz, and Horng (2024) and T.-Y. Lin et al. (2014), its brute-force processing of all spatial patches limits its efficiency and scalability.

Similarly, Marhuenda et al. (2025) model utilizes a Swin Transformer backbone combined with an Image Differentiation Embedding (IDE) to differentiate the temporal scans. While their model utilizes a lightweight 3-layer decoder, it still fuses and processes the full visual feature maps via cross-attention to generate answers. Other paradigms, such as Yung et al. (2024), attempt to bypass heavy paired-image processing by using Retrieval-Augmented Generation (RAG) to mix-and-match retrieved region features from an external database. While innovative, it requires a continuous, computationally expensive database search for every query and struggles to reason over novel differences not present in the retrieval set.

2.3 Mitigating Hallucination and Dataset Biases through Counterfactual Training

A persistent weakness in generative medical vision-language models is their propensity to hallucinate findings or rely on spurious dataset correlations rather than true visual evidence. Generative Diff-VQA models are particularly susceptible to this, as they must predict complex sequences describing evolving pathologies. Error analyses of current state-of-the-art models expose the clinical dangers of these hallucinations. For instance, evaluations of Cho et al. (2024) demonstrate instances where the architecture accurately identifies regions of interest but hallucinates the severity sequence, describing an abnormality in reverse temporal order. Similarly, researchers evaluating Marhuenda et al. (2025) noted a critical divergence between automated Natural Language Generation (NLG) metrics and clinical correctness. It frequently achieved near-perfect scores on metrics like Papineni, Roukos, Ward, and Zhu (2002) and ROUGE-L, yet clinical review revealed the model was misinterpreting diagnoses—often hallucinating the resolution of anomalies that had not actually subsided, or identifying new anomalies that were not present in the scans. This confirms that standard generative training heavily bi-

ases models toward the linguistic distribution of the training reports rather than grounded visual reasoning.

2.4 Hypothesis and Positioning

Existing methods indiscriminately process background and irrelevant features, diluting the model’s focus on the actual clinical question. This thesis hypothesizes that longitudinal, difference-focused questions can be accurately answered using only a handful of question-guided difference tokens. By introducing a Question-Guided Difference Tokenizer (QDT) that uses cross-attention to score and select only the Top-K most query-relevant visual residual tokens, the proposed model will explicitly filter out irrelevant changes, drastically reducing the computational load for the downstream language model while preventing the dilution of clinically salient deviations.

The limitations of direct subtraction and rigid graph mapping present a critical gap in modeling the direction and accuracy of temporal changes. This research hypothesizes that a Cross-Image Difference Attention (CIDA) module can overcome these limitations by semantically reconstructing and aligning prior-visit features to the current scan’s spatial layout via cross-attention before computing residuals. This alignment-guided approach is expected to provide a highly robust mechanism for distinguishing the true direction of change without being derailed by standard acquisition misalignments.

Standard cross-entropy optimization allows models to memorize language priors, leading to hallucinations on unchanged or subtly different image pairs. This research hypothesizes that counterfactual training, specifically combining hierarchical Kullback-Leibler (KL) divergence with Noise Contrastive Estimation (NCE), will force the model to explicitly disentangle true causal temporal evolution from nuisance visual patterns and dataset biases. By contrasting positive longitudinal pairs against logically opposite or synthetically perturbed counterfactual samples, the model is expected to significantly reduce its false positive hallucination rate on unchanged image pairs.

2.5 Comparative Analysis of Methodologies

The following table summarizes the key limitations of the current state-of-the-art models.

Table 2.1*Comparative Analysis of Limitations in State-of-the-Art DiffVQA Models.*

Model/Methodology	Computational Inefficiency	Spatial Misalignment Vulnerability Reasoning	High Data Dependency	Lacks Explicit Robustness Training
Hu et al. (2023)	✓	✓		✓
Lu et al. (2024)	✓	✓		✓
Yung et al. (2024)	✓		✓	✓
Cho et al. (2024)	✓	✓	✓	✓
Marhuenda et al. (2025)	✓	✓		✓
Our Proposed Model	<i>Addressed by QDT</i>	<i>Addressed by CIDA</i>	<i>Addressed by MRM</i>	<i>Addressed by Counterfactual Learning</i>

Table 2.2*Mapping of Proposed Components to Research Gaps*

Proposed Component	Research Gap Addressed	Corresponding Research Question
Question-Guided Difference Tokenizer (QDT)	Computational inefficiency of processing dense visual features.	RQ1: Can longitudinal questions be answered using only a handful of question-guided difference tokens?
Directional Residual Stack (DRS)	Vulnerability to spatial misalignments and implicit assumptions of direct feature substraction	RQ2: Does cross-image difference attention help the model distinguish the direction of change?
Counterfactual Regularizer (CFA-Reg)	vulnerability of generative models to clinical hallucination and dataset biases	RQ3: Can counterfactual training reduce the tendency of the model to hallucinate?

CHAPTER 3

METHODOLOGY

3.1 Architectural Overview

Longitudinal medical image comparison requires models to distinguish true pathological evolution from stochastic acquisition variability, such as patient movement or differences in posture. Conventional Difference-aware Medical VQA methods rely on direct feature subtraction, which implicitly assumes perfect spatial correspondence between scans. This thesis introduces ALIGN-VQA, an alignment-guided residual modeling architecture designed to overcome these spatial misalignment vulnerabilities.

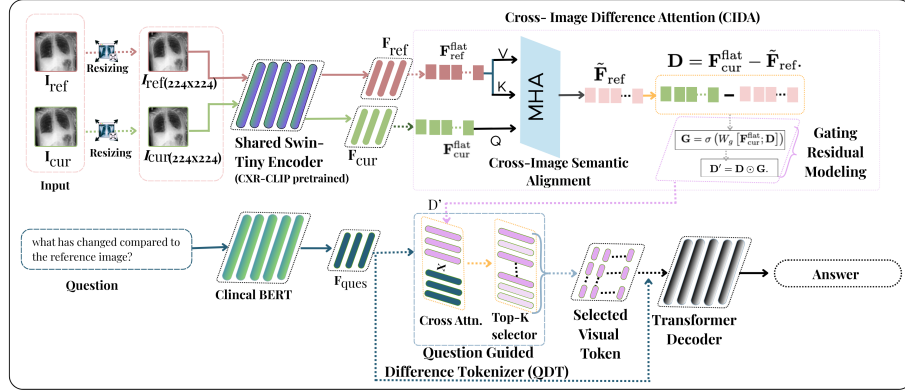


Figure 3.1

High Level Architecture Overview

As illustrated in Figure 3.1, the proposed methodology shifts from a brute-force comparison paradigm to an efficient, explicit reasoning framework. The architecture is composed of three primary functional blocks. Cross-Image Difference Attention (CIDA) and gated residual modeling semantically reconstructs and aligns historical features to the current scan before isolating evolution-aware deviations. Question-Guided Difference Tokenizer (QDT) employs cross-attention to selectively pool a sparse, highly relevant set of temporal changes based on the clinical query. Conditional answer generation and multi-stage training utilizes a lightweight transformer decoder optimized through a highly structured multi-stage training strategy to ensure grounded, hallucination-free reasoning.

3.2 Problem Formulation

Given a patient’s historical reference X-ray I_{ref} , a current follow-up X-ray I_{cur} , and a clinically grounded question Q , the objective of the Diff-VQA system is to generate an accurate natural language answer $\hat{A}$ describing the temporal evolution of the targeted pathology. Instead of mapping inputs directly to a dense difference map, our formulation isolates question-relevant residual deviations to conditionally autoregress the answer.

3.3 Architectural Components

3.3.1 Visual Feature Extraction

The visual processing pipeline begins by encoding the paired longitudinal images. Both I_{ref} and I_{cur} are resized to 224×224 and passed through a shared tiny variant of Liu et al. (2021). The shifted-window self-attention mechanism within the encoder enables highly efficient local-to-global context modeling. To inject domain-specific radiographic priors immediately into the feature extraction phase, the backbone is initialized with pretrained weights from You et al. (2023). This process yields two feature maps, which are flattened into token sequences:

$$F_{ref}^{flat}, F_{cur}^{flat} \in \mathbb{R}^{B \times N \times C} \quad (3.1)$$

where B is the batch size, C is the channel dimension, and $N = H \times W$ represents the number of spatial patches.

3.3.2 Cross Image Difference Attention

To compensate for spatial misalignments resulting from varying patient positioning and inspiratory effort between visits, we introduce the CIDA module. It utilizes Multi-Head Cross-Attention (MHA) to semantically align the historical features to the coordinate space of the current image. Specifically, the current image features are utilized as queries (Q), while the reference image features serve as keys (K) and values (V):

$$\tilde{F}_{ref} = \text{MHA}(F_{cur}^{flat}, F_{ref}^{flat}, F_{ref}^{flat}) \quad (3.2)$$

This operation reconstructs an aligned historical representation ($\tilde{F}_{ref}$) that is spatially coherent with the current scan, creating a reliable foundation for calculating true temporal differences.

3.3.3 Learnable Gated Residual Modeling

Once the features are aligned, the raw residual deviation is computed via element-wise subtraction:

$$D = F_{cur}^{flat} - \tilde{F}_{ref} \quad (3.3)$$

Because medical imaging inherently contains acquisition noise, utilizing the raw residual D directly can mislead downstream modules. Therefore, a learnable gating mechanism is applied. The gate calculates a channel-wise importance filter by concatenating the current features with the raw residual and passing them through a projection matrix W_g and a sigmoid function σ :

$$G = \sigma(W_g[F_{cur}^{flat}; D]) \quad (3.4)$$

The refined, evolution-aware residual features are then extracted via element-wise multiplication:

$$D' = D \odot G \quad (3.5)$$

The resulting tokens $\{D'_i\}_{i=1}^N$ encode clean, structured temporal changes.

3.3.4 Domain-Aware Text Encoding

The clinical question Q is encoded using Alsentzer et al. (2019), a language model extensively pretrained on clinical notes. This provides a domain-aware question embedding vector $F_{ques} \in \mathbb{R}^C$ that captures the semantic nuance of radiological terminology such as distinguishing between consolidation and atelectasis.

3.3.5 Question-Guided Difference Tokenization

Standard Diff-VQA models process all spatial tokens, which dilutes the signal of localized pathologies. The QDT module solves this by selectively pooling only the temporal changes that are relevant to the specific clinical question. Attention weights (α_i) are computed between the question embedding F_{ques} and each visual residual token D'_i using dot-product attention. To enforce extreme token sparsity and computational efficiency, a hard Top-K selection mechanism is applied:

$$T_{vis} = \text{TopK}(\{\alpha_i D'_i\}) \quad (3.6)$$

Based on empirical ablations, K is set to 16, ensuring the model forwards only the most highly discriminative visual evidence to the language decoder.

3.3.6 Conditional Answer Generation

The final response generation relies on a lightweight standard transformer decoder. The selected sparse visual tokens T_{vis} are concatenated with the question embedding F_{ques} . The decoder autoregressively predicts the answer $\hat{A}$ step-by-step, conditioned strictly on the question semantics and the isolated, aligned temporal changes.

3.3.7 Counterfactual Regularizer

To mitigate the model’s tendency to hallucinate, a counterfactual regularizer (CFA-Reg) is introduced. For each training sample (I_{cur}, I_{ref}, Q, A) , we generate a logical opposite by either swapping the images (I_{ref}, I_{cur}) or negating a directional term in the question (e.g., "worsened" $\rightarrow$ "improved"). The model performs a forward pass on both the real and counterfactual inputs, generating two distinct answer distributions, P_{real} and P_{cofa} .

3.3.8 Masked Residual Modeling

The masked residual modeling (MRM) process is designed to force the model to learn the contextual structure of the difference maps. First, an unlabeled image pair (I_{cur}, I_{ref}) is passed through the vision encoder to generate the concatenated residual map, R . A significant portion of the patches in this map are then randomly masked. The unmasked patches are fed into the QDT’s vision backbone, which is tasked with predicting the content of the masked patches. Figure 3.2 shows architecture of masked residual modeling module.

The training of MRM is guided by a reconstruction loss, which measures the dissimilarity between the model’s predicted patches and the original, unmasked patches. We use a Mean Squared Error (MSE) loss for this objective.

$$L_{\text{MRM}} = \frac{1}{|M|} \sum_{i \in M} (R_i^* - \hat{R}_i^*)^2$$

Here, M is the set of masked patch indices, R_i is the original patch, and $\hat{R}_i$ is the reconstructed patch.

3.4 Multi-Stage Training Strategy

The model is optimized using a comprehensive four-stage training regimen to ensure model robustness, prevent shortcut learning, and explicitly ground the vision-language

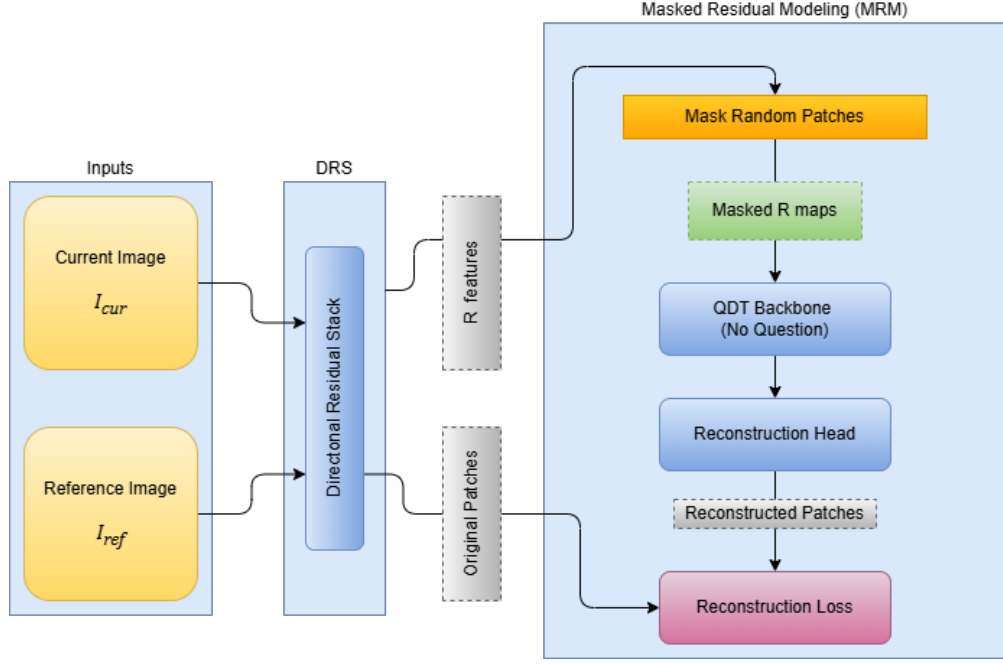


Figure 3.2

Masked Residual Modeling

alignment,

To enforce contextual coherence in the visual representation, 50 percent of the residual tokens are randomly masked in the first stage. A lightweight reconstructor is trained to predict the masked tokens from the visible context using a Mean Squared Error (MSE) objective ($\mathcal{L}_{mrm}$).

In second stage, a multi-label classification head is attached to predict 14 standard thoracic diseases from both the reference and current features using a Binary Cross-Entropy loss ($\mathcal{L}_{cls}$) to explicitly ground the visual features in known pathology

To mitigate spurious correlations and dataset biases and hallucinations, the model applies a combination of hierarchical Kullback-Leibler (KL) divergence and Noise Contrastive Estimation (NCE) in third stage. By contrasting true longitudinal pairs against synthetically perturbed counterfactual pairs, the model disentangles true temporal evolution from visual noise ($\mathcal{L}_{cf}$).

In the last stage, the entire architecture is fine-tuned for the generation task using a standard language cross-entropy loss ($\mathcal{L}_{lang}$) alongside an L1 sparsity regularization penalty on the gating weights ($\mathcal{L}_{gate}$) to force the gate to actively suppress non-essential in-

formation. The total loss of the complete training objective is weighted sum of these components.

$$\mathcal{L}_{total} = \lambda_{lang}\mathcal{L}_{lang} + \lambda_{gate}\mathcal{L}_{gate} + \lambda_{mrm}\mathcal{L}_{mrm} + \lambda_{cls}\mathcal{L}_{cls} + \lambda_{cf}\mathcal{L}_{cf} \quad (3.7)$$

3.5 Implementation Details

To ensure full reproducibility, ALIGN-VQA was developed using Paszke et al. (2019). The model features a highly efficient parameter footprint of approximately 182.5M parameters. The QDT module retains a maximum of 64 tokens during early training, which is dynamically pruned to $K = 16$ during fine-tuning. The maximum generated answer length is constrained to 128 tokens. Optimization is performed using the AdamW optimizer with a batch size of 128 and an initial learning rate of 5×10^{-5} , governed by a cosine annealing learning rate scheduler. The training schedule consists of 1 epoch of MRM, 3 warm-up epochs, and 20 epochs of end-to-end VQA fine-tuning. The empirically derived loss weights are set to $\lambda_{lang} = 1.0$, $\lambda_{mrm} = 0.1$, $\lambda_{cf} = 0.1$, and $\lambda_{cls} = 0.5$. Vocabulary mismatch issues during inference are handled dynamically by matching the initialized target vocabulary dimension to the exact embedding size preserved in the saved model checkpoints.

CHAPTER 4

EXPERIMENTAL RESULTS

4.1 Rationale

The primary objective of these experiments is to rigorously evaluate the clinical viability, computational efficiency, and reasoning accuracy of the proposed ALIGN-VQA architecture. Traditional Medical Visual Question Answering (Med-VQA) models often falter when applied to longitudinal tasks because they are vulnerable to spatial misalignments and tend to hallucinate clinical findings based on language priors. Therefore, the experimental design is motivated by three key goals. First, the author argues that spatial misalignments between historical and current scans are a major source of noise, and thus a Cross-Image Difference Attention (CIDA) module is necessary to establish a reliable baseline for comparison. Second, the author hypothesizes that since the vast majority of pixels in a longitudinal chest X-ray remain unchanged, selecting a sparse set of highly relevant tokens via Question Guided Difference Tokenizer (QDT) module will reduce the model’s tendency to hallucinate changes in stable anatomical regions. Finally, the author focuses on causal grounding through the use of counterfactual regularization (CFA-Reg). The author expects that by forcing the model to distinguish between true pathological evolution and dataset biases, penalizing it when it fails to flip its reasoning for a reversed image pair, the model will achieve more robust and hallucination-free clinical reporting.

4.2 Dataset

To test these hypotheses, the author utilizes Hu et al. (2025), which is a specialized derivative of the large-scale Johnson, Pollard, et al. (2024). Each sample in this dataset is structured as a longitudinal pair consisting of a historical reference radiograph and a more recent follow-up scan, stored in Digital Imaging and Communications in Medicine (DICOM) format. The x-rays images in Joint Photographic Experts Group (JPEG) format are fetched from Johnson, Lungren, et al. (2024) for faster preprocessing and training. These pairs are accompanied by clinically grounded questions that specifically target temporal changes, such as the progression of pleural effusion or the placement of medical devices. The dataset comprises 164,324 pairs of main and reference chest X-ray images, yielding a total of 700,703 question-answer pairs. This high volume of

data allows the model to learn the subtle semantic nuances required to distinguish between common clinical conditions like consolidation, edema, and atelectasis within a comparative context. Following standard benchmarking protocols, the dataset is strictly partitioned into training, validation, and testing sets using an 8:1:1 ratio. The Figure shows the distribution of seven distinct clinical categories

4.2.1 Pathological Distribution

Figure 4.1 illustrates the prevalence of medical findings across the dataset. The distribution exhibits a heavy long-tail characteristic typical of medical datasets. "No Finding" and "Support Devices" are the most prevalent positive labels, reflecting the high frequency of routine exams and ICU patient monitoring in Johnson, Pollard, et al. (2024). Among true pathological findings, "Pleural Effusion," "Lung Opacity," and "Atelectasis" occur most frequently. Conversely, critical but less frequent findings like "Fracture" and "Pleural Other" appear significantly less often. This class imbalance highlights the necessity for robust representation learning, particularly the Counterfactual Regularizer in the ALIGN-VQA architecture, to prevent the model from defaulting to high-frequency predictions.

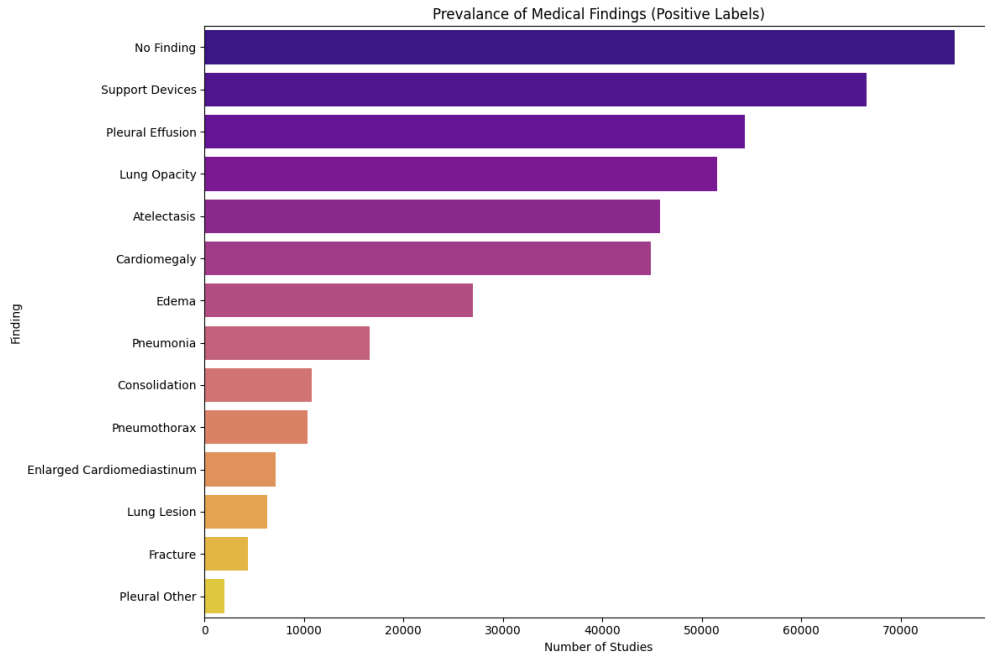


Figure 4.1

Prevalance of Medical Findings

4.2.2 Question Type Distribution

The questions are categorized into seven distinct clinical types, as shown in Figure 4.2. The distribution is relatively balanced among the top categories, ensuring a diverse evaluation of the model’s capabilities. The "Difference" category is the most frequent accounting for approximately 23 percent of the dataset, followed closely by "Presence", 22 percent and "Abnormality" 21 percent. The remaining categories—Location, Level, View, and Type—make up the rest of the dataset. While the ALIGN-VQA model is trained and evaluated across all seven categories, the primary focus of these experiments remains the "Difference" subset, as it explicitly tests the model’s ability to reason about longitudinal spatial and pathological changes between the reference and current scans.

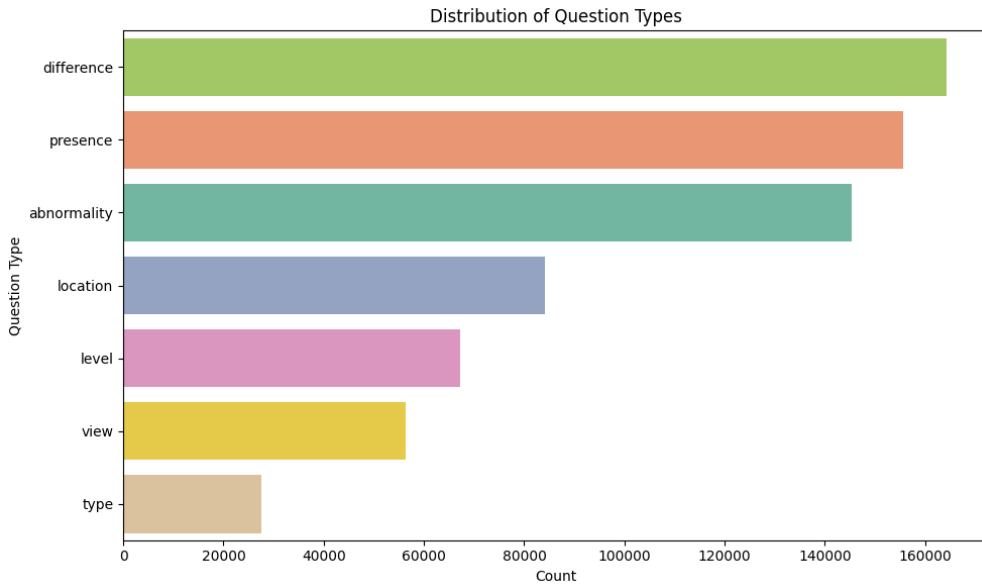


Figure 4.2

Distribution of Question Types

4.2.3 Linguistic Distribution

Figure 4.3 visualizes the distribution of question and answer lengths in characters. The question lengths exhibit a highly structured, multi-modal distribution with distinct peaks around 30, 42, and 50 characters. This rigid structure suggests that the questions were generated using a programmatic or templated approach based on the original radiology reports, rather than free-form human querying. The answer lengths are heavily skewed, with the vast majority of answers being extremely concise, under 25 characters, corresponding to binary "yes/no" answers or short categorical responses. A smaller, secondary distribution of longer answers, 50 to 150 characters, corresponds to the complex,

descriptive answers required for the "Difference" and "Abnormality" question types.

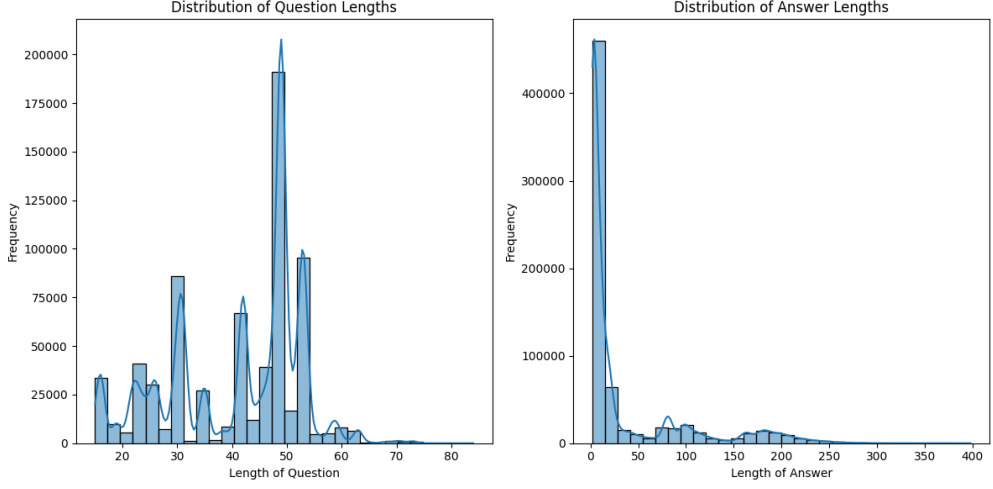


Figure 4.3

Distribution of Question Lengths

4.3 Experiment Setting

4.3.1 Preprocessing

To ensure standardized inputs for the vision-language pipeline, rigorous preprocessing protocols were applied to both the imaging and textual data. The resizing pipeline is engineered to maximize computational efficiency while preserving critical clinical details. Both historical and current chest X-rays were resized to 224×224 pixels to match the input dimensions expected by the tiny variant of Liu et al. (2021). Standard normalization was applied using You et al. (2023) pretraining statistics to ensure stable feature extraction. Clinical questions were tokenized using Alsentzer et al. (2019), with sequences padded and truncated to a maximum length of 48 tokens. To prevent index-out-of-bounds errors and size mismatches during evaluation, a common issue when dataset splits yield varying unique word counts, the inference scripts were programmed to dynamically extract the exact expected vocabulary size directly from the saved model checkpoints prior to initializing the classification head.

4.3.2 Models and Techniques

To benchmark the proposed ALIGN-VQA architecture, its performance is compared against several state-of-the-art models that encompass both general domain change-captioning and specialized medical Diff-VQA. Among the specialized medical models, the author evaluates Hu et al. (2023), a foundational Diff-VQA framework that relies on pre-extracted bounding boxes and an anatomical knowledge graph to model changes.

The author also compares the method with Cho et al. (2024), a computationally intensive transformer framework that utilizes massive external pretraining on natural images and radiology reports to process dense, unpruned feature concatenations. Additionally, the author benchmarks against Yung et al. (2024), a Retrieval-Augmented Generation (RAG) approach that generates pseudo-difference descriptions by mixing and matching retrieved regions from an external database. Furthermore, the author includes two recent generative medical VQA architectures including Lu et al. (2024), which employs a dedicated residual encoder and residual feature alignment to capture image disparities, and Marhuenda et al. (2025), which utilizes Liu et al. (2021) coupled with image differentiation embeddings to distinguish temporal scans prior to decoding.

Against these baselines, the author evaluate the proposed ALIGN-VQA, an alignment-guided residual modeling architecture featuring CIDA, a learnable gating mechanism, and the QDT. Beyond direct comparisons, extensive ablation studies are conducted on ALIGN-VQA by systematically toggling key components. For instance, the CIDA module is disabled to force direct feature subtraction, and the QDT Top-K selection threshold is altered. This allows the author to rigorously isolate and quantify the individual contributions of these novel mechanisms to the overall diagnostic accuracy.

4.3.3 Training Procedure

The model was implemented in Paszke et al. (2019) and trained end-to-end to ensure full reproducibility. All training procedures were executed on NVIDIA RTX6000 to efficiently accommodate the required batch sizes and the computational overhead of the attention matrix operations. The network was optimized using the AdamW optimizer with a batch size of 128. To ensure smooth convergence, the initial learning rate was set to 5×10^{-5} and governed by a cosine annealing scheduler.

The training process strictly adhered to a multi-stage strategy designed to incrementally build the model’s reasoning capabilities. This schedule commenced with one initial epoch of self-supervised Masked Residual Modeling (MRM), followed by three warm-up epochs, and concluded with 20 epochs of end-to-end Diff-VQA fine-tuning. The compound loss function was balanced using empirically derived weights, applying a weight of 1.0 to the language generation loss, 0.1 to the MRM loss, 0.1 to the counterfactual loss, and 0.5 to the auxiliary classification loss. Finally, early stopping was implemented based on validation CIDEr scores to continually monitor generalization

and prevent overfitting during the fine-tuning phase.

4.3.4 Evaluation Metrics

Because automated metrics often struggle to capture the clinical nuance of medical texts, the evaluation strategy utilizes a robust mix of strict string-matching, semantic similarity, and human validation. For automated assessment, the author reports standard Natural Language Generation (NLG) metrics, including Papineni et al. (2002), Banerjee and Lavie (2005), and C.-Y. Lin (2004), to evaluate sequence overlap and structural fluency. The author places a heavy priority on the Oliveira dos Santos, Colombini, and Avila (2021) metric, as its weighting mechanism explicitly penalizes generic answers and rewards the accurate generation of rare, clinically vital keywords, such as specific pathology names.

To move beyond the limitations of strict Exact Match (EM) criteria, a comprehensive manual qualitative evaluation of 1,384 test samples was also conducted. In this assessment, predictions were categorized by medical professionals as Fully Correct, Partially Correct or Incorrect. This manual review specifically focused on identifying any dangerous localization failures or clinical hallucinations, ensuring that the model’s outputs are not only linguistically fluent but also clinically safe and grounded in the visual evidence.

4.4 Quantitative Results on the MIMIC-Diff-VQA Benchmark

Table 5.1 presents the quantitative performance of the proposed ALIGN-VQA against current state-of-the-art models on the Hu et al. (2025). The evaluated baselines include early graph-based methods, retrieval-augmented models, dense-feature frameworks, and residual-based generative models.

As the results indicate, while models like Marhuenda et al. (2025) and Lu et al. (2024) achieve marginally higher scores on strict n-gram overlap metrics, the proposed model establishes a new state-of-the-art on the most clinically vital metrics: Banerjee and Lavie (2005) (0.480), C.-Y. Lin (2004) (0.739), and Oliveira dos Santos et al. (2021) (2.427). This distinction is critical. Papineni et al. (2002) uniformly weights all words, heavily rewarding the generation of common syntax templates. In contrast, Oliveira dos Santos et al. (2021) penalizes generic, high-frequency phrases while heavily rewarding the accurate prediction of rare, informative keywords such as specific pathology names or directional clinical terms. ALIGN-VQA’s dominant Oliveira dos Santos et al. (2021) score

of 2.427—outperforming the highly competitive Lu et al. (2024) (2.409) and vastly exceeding dense-feature models like Marhuenda et al. (2025) (2.119) and Cho et al. (2024) (1.832)—demonstrates that the extreme, QDT successfully filters out background noise without losing crucial diagnostic indicators.

4.5 Ablation Studies and Component Analysis

To isolate the contributions of the core architectural innovations, the author conducted extensive ablation studies, the results of which are detailed in Table 5.2. We evaluated the model by systematically toggling the CIDA module, CF-Reg, and varying the Top- k threshold in the QDT.

4.5.1 Impact of Architectural Modules

The baseline configuration without CIDA and CF-Reg performs poorly, yielding Oliveira dos Santos et al. (2021) of just 1.213. Introducing the CF-Reg alone improves the metric to 1.956, confirming its role in mitigating hallucinations. Conversely, introducing CIDA alone (Row 4) significantly boosts the metric to 2.172, providing empirical proof that explicitly resolving spatial misalignment before calculating residuals is far superior to direct feature subtraction. Combining both modules yields the optimal the score of 2.427, demonstrating that robust spatial alignment and causal, hallucination-free reasoning are highly complementary.

4.5.2 Impact of Token Sparsity

The author also investigated the effect of the token selection threshold, k , within the QDT module. Setting k too low ($k = 12$) limits the visual evidence available to the language decoder, dropping Oliveira dos Santos et al. (2021) to 2.258. Conversely, increasing the threshold ($k = 49$) incorporates excessive visual noise and irrelevant anatomical changes, diluting the clinical signal and lowering the metric to 2.226. The empirical sweet spot of $k = 16$ ensures that the model remains highly focused on the exact pathological deviations referenced by the clinical question.

Table 4.1

Quantitative performance comparison of the proposed model against state-of-the-art methods on the MIMIC-Diff-VQA dataset.

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr
Hu et al. (2023)	0.624	0.541	0.477	0.422	0.337	0.645	1.893
Yung et al. (2024)	0.705	0.633	0.572	0.517	0.381	0.651	1.804
Cho et al. (2024)	0.704	0.633	0.575	0.520	0.381	0.653	1.832
Marhuenda et al. (2025)	0.716	0.647	0.590	0.537	0.389	0.670	2.119
Lu et al. (2024)	0.710	0.636	0.580	0.530	0.395	0.736	2.409
Ours	0.653	0.578	0.518	0.466	0.480	0.739	2.427

Table 4.2

Ablation studies isolating the impact of the Top-k selection threshold, Cross-Image Difference Attention (CIDA), and Counterfactual (CF) regularizer.

Top- k	CIDA	CF	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr
12	✓	✓	0.636	0.541	0.502	0.450	0.481	0.700	2.258
49	✓	✓	0.631	0.555	0.496	0.445	0.480	0.713	2.226
16	×	×	0.399	0.340	0.296	0.256	0.274	0.485	1.213
16	✓	×	0.643	0.565	0.504	0.453	0.457	0.660	2.172
16	×	✓	0.661	0.577	0.511	0.453	0.454	0.661	1.956
16	✓	✓	0.653	0.578	0.518	0.466	0.480	0.739	2.427

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APPENDICES

VITA

This section presents a short description of the educational and professional achievements of the student.